NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-18950(NOrd)
23 January 1956
SUPERSEDING
NAVORD OS 1388
14 July 1952

PERFORMANCE SPECIFICATION

PROJECTILES, HIGH-CAPACITY, ANTI-AIRCRAFT, AND ILLUMINATING

1. SCOPE

1.1 Scope. This specification spplies to high-capacity, anti-aircraft, and illuminating projectiles, inert parts, and is augmented by drawings and specifications shown on the list of drawings and specifications shown on the list of drawings and specifications particular to the item of the contract.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-P-16594(BUORD) Projectiles, Bombs, Rocket, and Guided Missile Warheads, Inert Parts, Manufacture of

MIL-P-18948(NOrd) Projectiles, Painting, Marking And Greasing (By Projectile Manufacturers)

MIL-P-18949(NOrd) Projectiles, Illuminating, Localized Heat Treatment of

NAVY DEPARTMENT

General Specifications for Inspection of Material, Appendix II (Metals)

STANDARDS

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

PUBLICATIONS

NAVY ORDNANCE PAMPHLETS

OP 400 General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment

FSC 1315

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General.- In addition to the requirements specified herein the requirements of Specification MIL-P-16594 shall apply.

3.1.1 Pre-production lot.- Unless otherwise specified the submission of a pre-production lot and the satisfactory compliance of this lot with the requirements of this specification shall be required before general production on the contract is approved. These projectiles shall be made by the same processes the contractor intends to use in the fulfillment of the contract. The marking for such projectiles shall be as for production lots except the letters "PL" shall be substituted for the lot number.

3.2 Material.- The material for the body and baseplug shall be high grade homogeneous steel produced and fabricated by approved processes. Sufficient top and bottom discard shall be taken from the ingot to insure that the blooms, billets, or bars from which the projectiles are made are free from piping, segregation, flake, rolling or forging tears or similar defects as disclosed by fracture or macroetch examination. When only one projectile body is made from a single ingot, the base shall be toward the bottom of the ingot.

3.2.1 Physical properties.- Physical properties shall be as listed on the applicable drawings or as required in the appropriate specifications and shall be uniform for all projectiles in any lot.

3.2.2 Chemical composition.- Unless otherwise specified or limited by applicable specifications the chemical composition of the projectile components shall be at the option of the contractor. Prior to manufacture of projectiles the contractor shall submit, for approval by the procuring activity, a statement of the composition of the steel he proposes to use in the manufacture of the body and base plug. This statement shall show the maximum and minimum percentages of alloying elements intentionally added or otherwise present in the steel. The chemical composition of all projectiles furnished under the contract shall be in accordance with this statement as approved by the procuring activity and an analysis of each melt of steel used shall be furnished to the Inspector. If not otherwise limited by an applicable material specification the phosphorous and sulfur content shall each be limited to 0.05 percent maximum as determined from ladle analysis.

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3.3 Heat treatment.- All components shall be given such uniform heat treatment as the manufacturer desires to produce material conforming to the requirements of the specification. Components produced by cold forming operations or subjected to any cold work after heat treatment shall be adequately stress relieved to assure freedom from susceptibility to cracking or subsequent distortion as a result of unrelieved stresses. Hollow forgings shall not be quenched in water from above the critical temperature. The requirements of MIL-P-18949 -(NOrd) govern the localized heat treatment required on illuminating projectiles.

3.4 Freedom from defects.- All components shall conform to the requirements of MIL-P-16594 and the applicable drawings.

3.5 Surface finish

3.5.1 Exterior.- All exterior surfaces shall have a smooth finish and shall be free from burrs, nicks, sharp edges, sharp corners, rough spots, or areas which fail to clean up. Bourrelets and other surfaces which make contact with the gun shall have a superior finish. Surface roughness shall not exceed the values specified on the applicable drawing.

3.5.2 Cavities.- All cavities shall have smooth surfaces, free from burrs, nicks, sharp edges, sharp corners, rough spots or areas which fail to clean up. Cavities with surfaces as forged or cold formed are acceptable provided all requirements of finish, concentricities, dimensions, etc., are met, see MIL-P-16594.

3.6 Shape.- The interior and exterior shape of the projectile as shown on the applicable drawing, shall be maintained.

3.7 Interchangeability.- Components such as fuze hole plugs, adapters, etc., which are not permanently assembled by the contractor or which because of the requirements of fit or other considerations are not clearly intended to be used in selective assembly, shall be entirely interchangeable.

3.8 Concentricity.- All requirements of concentricity specified on the applicable drawings or specifications shall be adhered to. The longitudinal axis of the cavity shall coincide with the longitudinal axis of the external contour within the limits specified on the applicable drawing (expressed as difference in wall thickness) or for 4" and larger projectiles by the roll test specified in 4.3.6.

3.9 Band and band seat.- The material for the rotating band shall be as specified on the applicable drawing. The band seat shall be roughened before banding as shown on the applicable drawing or by a method approved by the procuring activity. For requirements governing the material, assembly, and inspection of projectile rotating bands see the applicable specification shown on the pertinent list of drawings.

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3.10 Ballistics.- The dispersion, ranges, and other behavior of projectiles shall not differ greatly from those already in service or those tested in the development stages. Projectiles when fired down the range shall have smooth flight and shall not break up, upset, strip their bands, or develop other faults which would affect their serviceability. In all cases, the dispersion shall not be such as to affect the value of the projectiles in service.

3.11 Marking

3.11.1 Process marking.- The contractor shall mark projectile bodies and other components as required by the Inspector to identify the component with its melt, ingot number, heat-treatment or other pertinent processes or stages of manufacture.

3.11.2 Identification marking.- Finished projectiles shall be marked in accordance with the applicable drawing and the specification indicated under 2.1. For projectiles assembled from bodies, base plugs or other components which are not entirely interchangeable, each component to be assembled together shall be marked with the same serial number.

3.12 Cleaning and preservation.- All surfaces of finished projectiles shall be cleaned and coated in accordance with the requirements of MIL- P-18948 (NOrd) specified in the contract.

3.13 Base plugs.- For projectiles with base plugs, the base plugs and their mating counterbores in the projectile bases shall be such that a gas tight fit is obtained when the base plug is assembled against its gasket. Base plugs for high-capacity and anti-aircraft projectiles shall be made in such a manner that the longitudinal axis of the plug is normal (90°) to the fiber flow lines or grain flow lines of the material of which the plug is made. When applicable, Acme threads of special form and fit may be used on the exterior of the base plugs; the threads shall be individually fitted into the projectile if necessary to obtain an acceptable fit, as close fitting threads with a minimum amount of clearance or shake are necessary, but the base plug shall be readily removable when received at the loading plant.

3.14 Gaskets.- When gaskets are specified to be supplied under the contract, they shall be in accordance with the applicable drawing and specification. Unless otherwise specified, gaskets shall not be assembled with the projectiles but shall accompany each lot of projectiles; the number supplied shall be 10 percent in excess of the number of projectiles.

3.15 Miscellaneous components.- The applicable list of drawings and specifications references various components utilized in the assembly of projectiles. It shall be the contractor's responsibility to acquaint himself with the detail requirements of such components specified under the contract.

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4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling.- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of MIL-STD-105.

4.1.1 Inspection lot.- As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot," i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.1.2 Production lot.- The number of projectiles in a production lot shall be as follows:

Caliber	Lot size
From 3" to and including 4"0 - - - - -	1000
Above 4" - - - - - - - - - - - - - - - -	500

The production lot size shall be exclusive of projectiles furnished for ballistic test or retest. At the discretion of the procuring activity, the production lot size may be increased if the uniformity and quality of tested lots indicate that a change is justified. Changes in lot size shall not operate to change the number of projectiles, including test projectiles, to be furnished under the contract.

4.1.3 Samples for ballistic tests.- Finished projectiles shall be submitted by production lots for selection by the Inspector of five samples from each lot for ballistic tests. Sufficient allowance for test projectiles shall be made for each lot. In general, the Inspector will give consideration to the complete history of manufacture of the lot and will select such projectiles for the ballistic tests as will demonstrate the uniformity and quality of the lot. However, no interpretation of this paragraph shall operate to prevent the selection of samples representing the least favorable quality as required by the General Specifications. At the discretion of the procuring activity, the Inspector may be authorized to ship lots prior to completion of the ballistic tests.

4.1.4 Samples for chemical analysis.- When no check of the chemical analysis is made by the government, the analysis furnished under 3.2.2 shall govern. Otherwise, samples shall be taken in accordance with Appendix II, General Specifications for Inspection of Material. When required by the Inspector, a check chemical analysis of any bar or forging of any melt shall be made to determine the uniformity of the material.

4.2 Inspection.- The general inspection procedures shall be in accordance with General Instructions for the Design, Manufacture, and Inspection of Naval

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Ordnance Equipment, OP 400; Government Inspection shall be in accordance with the provisions of General Procedure for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.3. Tests at Contractor's plant.- In addition to the special tests described herein, the Inspector may require other tests, measurements, etc., necessary to insure that the projectiles are in accordance with the applicable specifications.

4.3.1 Fracture and macro-etch tests.- The Inspector shall ascertain that sufficient discard has been made from each ingot to insure freedom from piping and segregation. This shall be determined by either a fracture test or a macro-etch test which shall be conducted on the cold stock representing the uppermost portion of each ingot. The fracture of macro-etched section shall show at least 10% of the cross-sectional area of the ingot, bloom, billet or bar; shall be at least 2" in diameter; shall be centrally located to show the axis of the ingot and shall be free from indications of piping or segregation. Discard shall be continued until the fracture or macro-etched section discloses satisfactory material. Should the Inspector require it, a fracture may be ground flat and suitably etched further to indicate the condition of the material.

4.3.2 Tension tests, projectiles.- Tension test specimens shall be taken from sample projectile bodies selected to represent each production lot. The choice of the number of test specimens taken from any sample unit and their locations in the sample unit shall lie with the Inspector. One or more samples from each lot shall be selected according to the following schedule:

a. If all projectiles in a lot are from the same melt and are heat-treated together, at least one sample shall be selected.

b. If all projectiles in a lot are from the same melt and are heat-treated in the same equipment, under identical conditions, and in uninterrupted sequence (whether processed individually or in batches) at least one sample shall be selected.

c. If projectiles in a lot are from more than one melt, at least one sample from each melt shall be selected, except for the following qualifications:

(1) In the case of large projectiles where more than two melts are required for one lot, the consideration of two or more similar melts as a single melt may be authorized by the Bureau if recommended by the Inspector who must be satisfied that the equipment and control thereof is adequate and is being operated in such a manner as to make such a procedure acceptable.

(2) In the case of mixed lots having a small percentage of projectiles from a melt that has been satisfactorily heat-treated as a previous lot in the same equipment, and under the same conditions the selection of another sample from this melt in the given lot may be waived at the discretion of the Inspector. Since the specifications require that samples shall be selected that represent the least favorable quality, it is to be expected that several time the minimum number of samples will need to be selected at the start of production. As production

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progresses the number of samples may be decreased but in no case shall fewer than the minimum number specified above be selected. The choice of the number of samples to be selected from any lot shall lie with the Inspector.

4.3.3 Tension tests, base plugs.- Base plugs, heat treated together and made from the same melt, shall be selected for and subjected to the tensile test in accordance with MIL-STD-105, code letter C, AQL 4.0%. The number of tension tests conducted on each unit shall be prescribed by the Inspector with a minimum of one per unit. Subsize test specimens, machined to the form and dimensions, of the largest obtainable specimen shown in Type 4, Appendix II, General Specifications for the Inspector of Material, may be used in the tension testing of base plugs, if and when necessary.

4.3.4 Hardness test.- Similarity in a group of projectiles from which representative tension tests are selected shall be determined on a sample selected in accordance with MIL-STD-105, and tested in accordance with Appendix II, General Specifications for the Inspection of Material.

4.3.5 Magnetic particle test.- When required by the Applicable drawing, projectiles shall be subject to magnetic particle inspection. The contractor may accomplish screening inspection of all projectiles using magnetic particle equipment capable of disclosing the presence of cracks with reliability as good as that possessed by the dry process method. All projectile bodies so tested and having definite indications of cracks shall be discarded by the contractor unless compliance with the drawing requirement is demonstrated by dry process magnetic particle inspection. Defect depths less than 0.01 inch may be distinguished from those exceeding 0.01 inch by measuring widths of magnetic particle deposits. For an authorized procedure see 6.6. Deviations from the procedure of 6.6 will be authorized provided the techniques demonstrate ability to yield equivalent results.

4.3.6 Concentricity test.- A projectile having a difference in wall thickness in localized areas which exceeds, but it not more than twice the value specified on the applicable drawing may be subjected to a roll test to determine satis-- factory balance. In the roll test, the projectile shall be rolled on the bourrelets along horizontal rails and any tendency to settle shall be noted. The projectile shall be considered to be satisfactorily balanced if any tendency to settle can be completely overcome when the application of a weight to the circumference at a point diametrically opposite the settling point, exerts sufficient torque to become the settling point after the projectile has been rotated to place the weight 90° from the top position. The following table lists the maximum acceptable torque that may be exerted (usually applied by means of a magnetized weight):

Caliber	Balancing Torque (inch ounces)
16"	328
14"	224
12"	168
8"	32
6"	21
5"	12½
4"	4

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4.4 Ballistic test.- After selection of the samples by the Inspector in accordance with 4.1.3, the test projectiles shall be prepared for shipment and delivered as directed by the procuring activity. Test projectiles inert loaded or loaded as for service use (Illuminating projectiles shall be loaded as for service use) at the discretion of the procuring activity, shall be fired down the range to determine compliance with 3.10. Upon satisfactory performance the lot will be accepted provided all projectiles in the lot are otherwise acceptable.

4.4.1 Ballistic retests.- If the performance of the flight test is unsatisfactory, the contractor may request additional tests. Additional tests, if authorized, will be made under conditions specified by the procuring activity. The submission of lots shall be suspended until satisfactory performance is obtained, unless otherwise directed by the procuring activity.

4.5 Pre-production lot test.- Projectiles in the pre-production lot will be subjected to all tests considered necessary by the procuring activity to assure compliance with the requirements of this specification. The results of these tests will determine whether the contractor shall proceed under the contract.

5. PREPARATION FOR DELIVERY

5.1 Preservation.- All surfaces shall be cleaned, preserved and greased as required. The paint on painted surfaces shall be thoroughly hardened and dried.

5.2 Packaging

5.2.1 Each projectile shall be packaged or fitted with suitable plugs to protect all threads and cavities.

5.2.2 The rotating bands shall be suitably protected from damage by the use of approved grommets or other devices.

5.3 Packaging

5.3.1 Projectiles shall be crated, boxed, or palletized in accordance with applicable specifications as specified in the contract. Such packaged units shall be reasonably protected while awaiting shipment.

5.3.2 Projectiles shall be loaded into such conveyance as the procuring activity may designate, unless it is agreed that loading may be delayed.

5.3.3 Spare parts, if required by the applicable specifications, shall be packed to accompany each lot.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Caliber, mark and mod, and date of applicable drawings.
(c) Applicable list of drawings number.
(d) Type of preservation and coating required.

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6.2 Termination of contractors responsibility.- The responsibility of the contractor for any projectile made under these specifications shall terminate two years from the date of delivery of that projectile.

6.3 Cost of conducting ballistic tests and retests.- The cost of conducting ballistic tests, except retests, will be borne by the Government. Unless otherwise specified costs of conducting ballistic retests will be borne by the contractor.

6.4 Parts furnished by the government.- In case new parts are added or substituted for those originally contracted for, the procuring activity reserves the right to furnish such parts. However, if required, the contractor shall assemble such parts in the specified manner and will be reimbursed for any extra expense entailed thereby, as provided for in the contract.

6.5 Precedence.- All correspondence, instructions, or practices that are in conflict with and are dated or in use prior to the latest revision of this specification are hereby cancelled.

6.6 Magnetic particle inspection of 3" projectiles.- Examination of the surface should be conducted with the axis of the projectile in a horizontal position. To facilitate handling, projectiles may be set on brass rollers. Powder should be dusted on the projectile and excess powder removed by use of two (2) pounds air pressure from a hose held three (3) inches from the surface of the projectile (one method of maintaining this distance is to attach a wire to the hose extending 2-3/4" from the end of the hose).

Any powder indication obtained by this method which equals or exceeds a measured width of 0.010 is evidence of a surface defect exceeding 0.001 in depth. (The width of accumulation of powder may readily be measured with a Brinell microscope having built in scale calibrated in thousandths of an inch).

6.6.1 Equipment.- The following equipment may be utilized:

(a) Model KMS-2DA magnetizing unit manufactured by Magnaflux Corporation.
(b) Fifty (50) foot length of oxygen hose (1/4" I.D.) attached to air line through low pressure (0-10 lb.) gage.
(c) Gray magnetic powder (magnaflux #1)
(d) Brinell microscope
(e) Narrow coil Solenoid Demagnetizer (Magnaflux Corporation Model SSB 1416).

6.6.2 Cleaning of projectiles.- Projectiles should be cleaned on outside surface and in cavity to remove oil or grease which would interfere with magnetization and application of powder.

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6.6.3 Magnetization - Projectiles should be magnetized residually to produce circular field by means of current passed through 5/8" to 3/4" diameter copper prods making contact through the base of the projectile. Projectile should be flash magnetized by momentary current of 400 to 500 amperes. The copper prod held against the base of the projectile should have the contact end machined flat. The contact end of the prod held in the cavity should be shaped to a 3/8" radius. Contact ends of the prods should be periodically dressed to minimize possibility of arcing.

6.6.4 Demagnetization.- Subsequent to magnetic particle inspection, the acceptable projectile shall be demagnetized by moving it through the coil of the demagnetizer. The projectile shall be placed in the coil close to the inner wall of the coil and the demagnetizing current left on until the projectile has been withdrawn several feet from the coil.

Patent notice.- When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0-8278

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